

Performance Evaluation of Tapped Delay Line Propagation Model in an Urban Environment

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Abstract

Propagation models are a useful tool for the design of both indoor and outdoor wireless systems, thanks to them it is possible to determine the probability that the signal reaches or not to a specific target as well as its power level at the receiving point.

This article presents the results of the Performance Evaluation of the tapped delay line propagation model in a dense urban environment in the City of Bogota-Colombia for an 854 MHz frequency. Model validation was performed from actual measurements at three different points of a major route. The results show that at the nearest point to the transmitter there is a lower average error between the measured data and those calculated by the model. The average error of the propagation model is approximately 3.58 dB.

Keywords: Propagation Model, Multipath Channel, Losses, Delay, Frequency.

Introduction

The main problem with wireless signal reception is fading caused by multipath propagation, coupled to other problems such as intersymbol interference (ISI), obstacles, interference, among others [1]. Therefore, knowledge of the radio channel is essential for deploying and developing a wireless system.

The tapped delay line (TDL) model is used to model channels when channel conditions are constantly changing due to position of the user and the different obstacles that appear at each position. This paper validates the performance and accuracy of the TDL model in a dense urban environment.

There have been several studies on TDL propagation models, in [2] and [3] model studies are presented in which many samples are taken in the 5GHz band to demonstrate that the TDL model is useful to design communications networks in indoor environments. On the other hand, [4], [5] and [6], it presents the behavior of the model in outdoor settings, validating it is effective in the design of wireless systems in the 3 GHz band. Particularly [7], within its conclusions it determines the number of taps to be used in each scenario based on the distance to the receiver. Moreover, [8] and [9] channels of this type on bio-sensors that are attached to body

tissues are modeled. In [10] and [11] state problems on the assumption of the Wide-Sense Stationary Uncorrelated Scattering (WSSUS) [12], which suggests that channel correlation between the response of a channel at different (t) times and (f) frequencies is minimal and therefore negligible, thus leading to the above condition, which is taken into account in virtually all propagation models.

[13] and [14] change some elements to propose theoretical model improvements to the model such as variation of sampling or to make discreet the channel that was initially modeled as continuous. The results are not definitive, but allow further research to confirm the results obtained in very specific scenarios. Finally, in [15] a TDL model for Code Division Multiple Access (CDMA) systems that reduces the complexity of the simulation is developed.

The rest of the paper is structured as follows. Section II introduces the definition and description of the TDL model for different signal delays is presented. Section III describes the simulation and experimentation setting for a dense urban environment at different distances. Section IV shows measurements and results achieved, and the TDL propagation model is validated. Section V draws the conclusions.

TDL Propagation Model

By taking multiple paths from the transmitter to the receiver, we obtain a multipath channel (MPC), these paths are generated by the fact that there is not only a line of sight but there are also delayed arriving signals, which cannot be ignored [16] due to reflections caused by fixed (buildings) and mobile (vehicles) obstacles. Hence, there are other paths that may be subject to reflections, shading or dispersion, and each one of these signals along the line of sight, reach the destination point at different times due to different advanced distances. A highly suitable method to model a channel characterized by multipath propagation is the TDL model [16], which is generally comprised by an N delay series with τ duration.

Among the N delays there are taps, each one of which is affected by an $h(t)$ channel response, usually modeled as zero-mean uncorrelated complex Gaussian processes. This uncorrelated condition, which does not occur in time or

frequency, is called WSSUS and was established in [12]. Since then it is taken into account in almost all models. The TDL model structure TDL with $N + 1$ taps can be seen in Fig.1. If the user is not moving it is said that the channel is time invariant; otherwise, it is said that the channel is time varying because the radio paths and their associated delays vary over time.

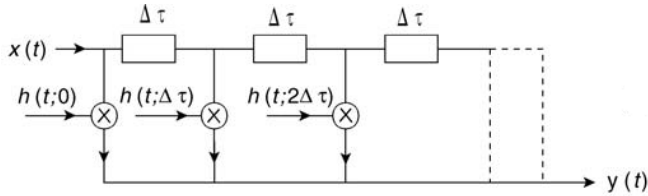


Fig.1. TDL Model with N+1 Taps (paths) [17].

A common assumption is that path changes occur at a constant rate depending on the angle of incidence with respect to the movement direction of incidence.

The response to a varying channel in time can be described as follows, in the case that there is no variation in time, $h(\tau)$ it is the channel response in a given t time to an impulse transmitted in advance of τ seconds. This response does not depend on t but instead only in the τ delay. When the channel is variable in the response $h(t, \tau)$, it depends on the observation time t and delay τ , hence the response is the one observed in t time, at the moment of applying an impulse in the $t - \tau$ moment. If this is particularized to a time invariant channel, we would have: $h(0, \tau) = h(\tau)$ [18] [19].

The description of a linear system in the time domain can be specified by its impulse response, as the channel varies over time, then the response will too. If the impulse response $h(t, \tau)$ is then the complex envelope in the receiver side $y(t)$ is related to the $x(t)$ input given by the convolution integral, as shown in Equation (1).

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau) * h(t, \tau) d\tau \quad (1)$$

where timing variations are contained in $h(t, \tau)$ fluctuations. This convolution can be rewritten as a summation, "see Equation (2)".

$$y(t) = \Delta\tau \sum_{m=1}^n x(t - m\Delta\tau) h(t, m\Delta\tau) \quad (2)$$

This expression provides a physical representation of the channel in the form of tapped delay line (TDL) composed of elements of differential delays and modulators. The channel can also be represented by the transfer function varying in time, $T(f, t)$ as shown in Equation (3).

$$y(t) = \int_{-\infty}^{\infty} X(f) T(f, t) \exp(j2\pi ft) df \quad (3)$$

where $T(f, t)$ function is the Fourier transform of $h(t, \tau)$ with respect to the variable τ .

Both the channel and the transmitted signal must be represented in the discrete time domain in order to carry out model simulations.

The sampling frequency (f_s) must consider the maximum Doppler deviation (f_m) and take into account the bandwidth (B) of the signal as well. Therefore, the required sampling frequency is represented by Equation (4).

$$f_s \geq 2(B + f_m) \quad (4)$$

The discrete time channel model assumed is transversal, that is, an FIR filter with variable taps over time given by $g_i(t)$ function and spaced by a time $\Delta\tau = t_s$ (see Fig.2).

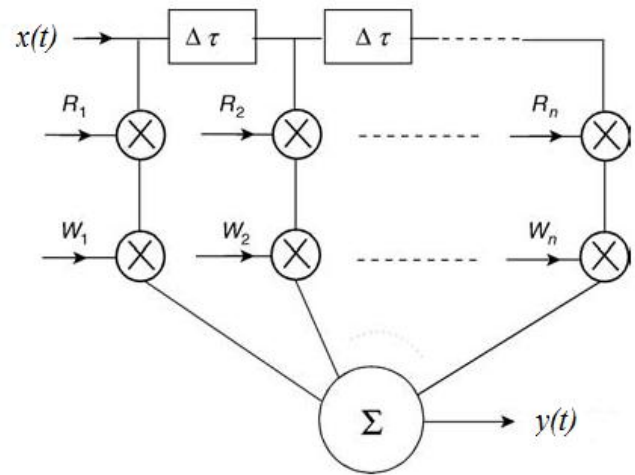


Fig.2. TDL Model for a Variant Channel in Time [19].

If the condition WSSUS is assumed, the complex $g_i(t)$ coefficients are uncorrelated and have Gaussian distribution. This model is a direct representation of the set of delay-Doppler probability distributions $p(\tau, \nu)$ which is proportional to the dispersion function $R_s(\tau, \nu)$. In the COST 207 model [20] this probability density is given by Equation (5).

$$p(\tau, \nu) = p(\tau) p(\nu / \tau) \quad (5)$$

That is to say, the channel is modeled in terms of a distribution of delays $p(\tau)$ and the Doppler distribution, taking each of the delays $p(\nu / \tau)$.

In Fig. 2, two elements are observed per tap R_i y W_i . The first is the Rayleigh distribution with a given Doppler

spectrum and the second corresponds to a weight that indicates the shared power in each tap. By combining these two elements into one, we obtain the variation coefficient in time $g_i(t)$.

This model makes a simplification (which was not done in the model COST 207) and assuming that the dispersion function $R_s(\tau, \nu)$ [19] and consequently also the function of joint probability density $p(\tau, \nu)$ are separable. This assumption simplifies the simulation because it is equivalent to say that the variables τ and ν are statistically independent, resulting in Equation (6).

$$p(\tau, \nu) = p(\tau)p(\nu) \quad (6)$$

The probability density function of delay and power is an exponential distribution, as described in Equation (7).

$$P(\tau) = a * \exp(-\tau / b) \quad (7)$$

Or a combination of two exponential for large delays scenarios corresponding to uneven terrain such as a dense urban environment.

Three types of Doppler spectra are then defined, according to signal delays: Classic Spectrum (Class), Gaussian Spectrum 1 (Gauss1) and Gaussian Spectrum 2 (Gauss2).

A. Classical Spectrum (Class)

It is used for delays less than 500ns and is described by Equation (8).

$$S(\tau_i; f) = \frac{A}{\sqrt{1 - (f / f_m)^2}} \quad \text{para } f \in (-f_m, f_m) \quad (8)$$

where, f_m is the maximum Doppler shift and A is used to normalize the filter that does not distort the average power. This corresponds to the so called Jakes/Clark filter.

B. Gaussian Spectrum 1 (Gauss1)

It is used for delays between 500ns and 2ms. It is given by the sum of two Gaussian functions as shown in Equation (9).

$$S(\tau_i; f) = G(A, -0.8f_m, 0.05f_m) + G(A, 0.4f_m, 0.1f_m) \quad (9)$$

where A_1 is 10dB.

C. Gaussian Spectrum 2 (Gauss2)

It is used for delays longer than 2ms. It is given by the sum of two Gaussian functions as shown in Equation (10).

$$S(\tau_i; f) = G(B, -0.7f_m, 0.1f_m) + G(B_1, 0.4f_m, 0.15f_m) \quad (10)$$

where B_1 it is equal to 15dB.

TDL Model Simulation

The classifications proposed in [20] for a "bad-urban" dense urban environment, are used for simulation of the TDL model at a frequency of 854 MHz. Table 1 shows the characteristics relating to the simulation of the urban scenario, using 12 taps. In this case the Doppler category is different and depends on distances.

TABLE.1. Doppler Categorization According to Data of the COST-207 [21] Model for Dense Urban Environments.

Tap #	Delay (μs)	Power (dB)	Doppler Category
1	0	-7	Class
2	0.2	-3	Class
3	0.4	-1	Class
4	0.8	0	Gauss1
5	1.6	-2	Gauss1
6	2.2	-6	Gauss2
7	3.2	-7	Gauss2
8	5.0	-1	Gauss2
9	6.0	-2	Gauss2
10	7.2	-7	Gauss2
11	8.2	-10	Gauss2
12	10.0	-15	Gauss2

The simulation results for the 12 taps are presented in Table 1, using the respective filters according to the Doppler category as shown in Fig.3.

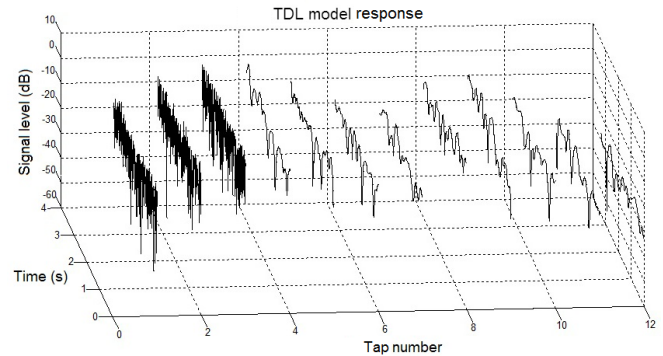


Fig.3. Simulation Model for 12 Taps at 854 MHz.

According to these results it is found that the received power varies depending on the filter used, a fact that shows the great importance of the Doppler Effect. Still, as expected, while the delay increases the power decreases, because the transmitter is at a greater distance.

TDL Model Validation

To validate the model, several measurements were performed on a test scenario that allows to make a comparison with the performed simulation. The chosen scenario corresponds to Carrera 7 (Avenue 7th), between streets 39th and 41st in the City of Bogota, Colombia. In Fig.4, it is observed the

reception infrastructure used for measuring signals, which is composed of an ultra-wideband antenna and a spectrum analyzer having the multi-path power module.

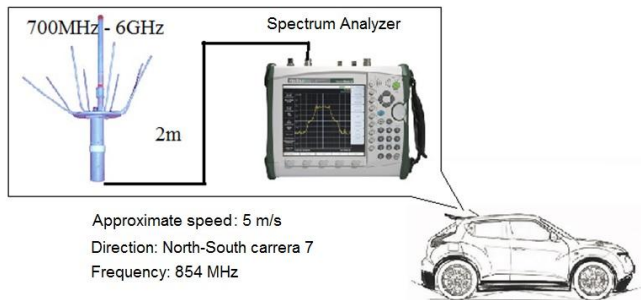


Fig.4. Receptor Infrastructure

Table 2 presents the characteristics of the transmitter (Tx), corresponding to a base cellular mobile station (BTS).

TABLE.2. Data Transmitter

Coordinates: Lat-4° 37' 43", Long-74° 3' 53", Height 2625m	
Band	850 MHz
Cell_Id	26251
Antenna Gain	17 dBi
Power Transmission (Min-Max)	30 dBm-45 dBm

Fig.5. shows three points A, B and C, where measurements were taken at Carrera 7, place where the vehicle that had the receiver and antenna analyzer was moving at an approximate speed of 5m/s (18km/h).

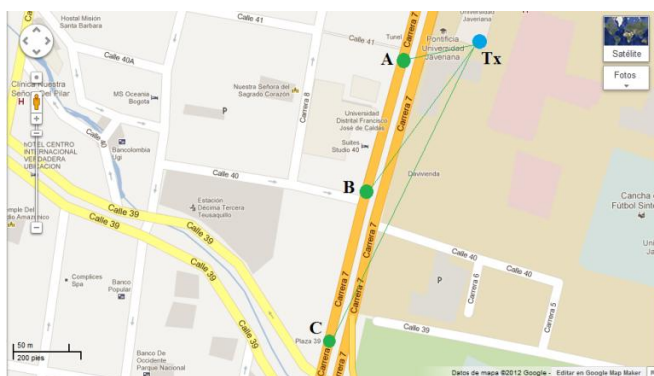


Fig.5. Dense Urban Setting where the Measurements were taken

The results of measurements and simulations are presented in Tables 3, 4 and 5. Here it is noted that in some cases at longer distances the received power is greater, this obeys not only to multipath signal propagation but also to power control exercised by the BTS, which involves increasing the transmission power of BTS as the receiver moves away. Table 6 shows a relationship between power means obtained in the simulation and measurements.

TABLE.3. Measurement at Point A in Fig. 5

Distance to BTS	91.7 m									
Delay	0.3 μ s									
Time	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	
Received Power (dBm)	-59	-51	-48	-47	-50	-48	-50	-57	-52	
Simulated Power (dBm)	-45	-50	-60	-55	-55	-50	-70	-46	-45	

TABLE.4. Measurement at Point B in Fig. 5

Distance to BTS	179.13 m									
Delay	0.6 μ s									
Time	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	
Received Power (dBm)	-42	-44	-43	-47	-42	-45	-46	-43	-44	
Simulated Power (dBm)	-35	-40	-40	-45	-34	-61	-51	-33	-28	

TABLE.5. Measurement at Point C in Fig. 5

Distance to BTS	310 m									
Delay	1.03 μ s									
Time	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	
Received Power (dBm)	-58	-49	-49	-48	-49	-47	-52	-48	-46	
Simulated Power (dBm)	-32	-45	-65	-45	-44	-42	-49	-36	-39	

TABLE.6. Comparison between Measured and Calculated Power with the TDL Model

Point	Simulated Average Power (dBm)	Measured Average Power (dBm)	Error (dB)
A	-52.88	-51.33	1.55
B	-40.77	-44.56	3.78
C	-44.11	-49.55	5.43

Fig.6 shows the simulation results for delays corresponding to each of the three points A, B and C.

The data obtained by simulation, are in general, close to the measured values; although there are some that affect the error. If the data of the three points are taken and averaged, it is found that the total error between theoretical and actual results is approximately 3.58 dB.

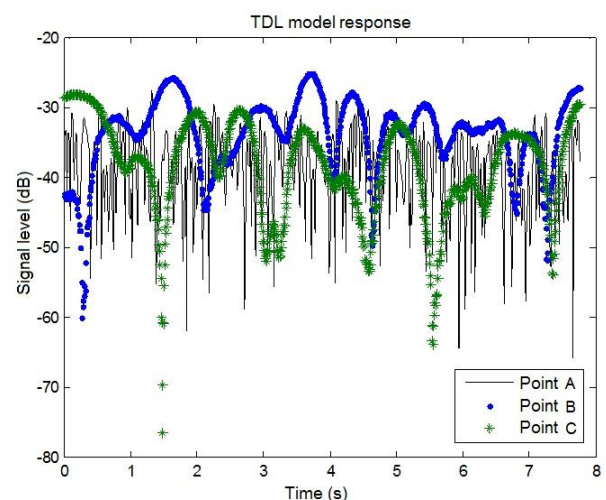


Fig.6. Results of the Power Received by the TDL Model

Conclusion

The results show that the model accuracy decreases as the receiver moves away from the BTS. At point A, the average error between simulated power and measure is the lowest, because there are fewer obstacles between the transmitter and the receiver, and therefore a better line of sight, which also results in fewer paths. At points B and C a significant increase in the error is observed, since in both cases there is the greatest influence of traffic environment as well as changes in trajectory and delays, since *Carrera 7* is highly trafficked by all kinds of vehicles which makes paths to have high variance and delays with abrupt changes at different times. This goes against one of the approaches of the model which indicates that path changes must occur at a constant rate.

It is clear that path variations or loss of the line of sight are reflected in the model output since response to impulse over time $h(t, \tau)$ is always given in terms of the observation time and the delay the signal experiences to reach the mobile at t , time, that is, the signal was transmitted in $t - \tau$, time and this τ it can be highly variable.

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